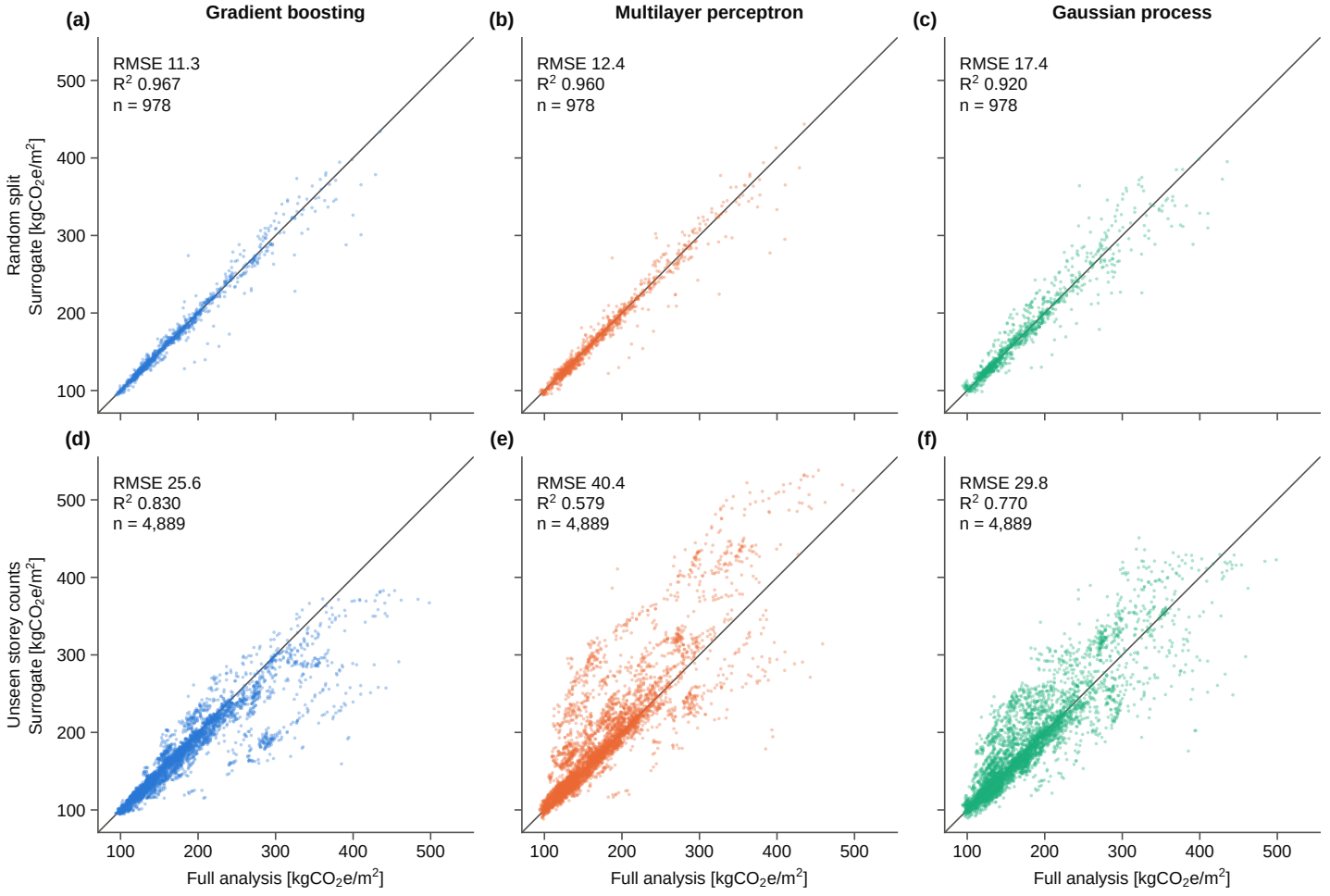

(a)
Gradient boosting
(b)
Multilayer perceptron
(c)
Gaussian process
RMSE 11.3
R2 0.967
n = 978
RMSE 12.4
R2 0.960
n = 978
RMSE 17.4
R2 0.920
n = 978
(d)
(e)
(f)
RMSE 25.6
R2 0.830
n = 4,889
RMSE 40.4
R2 0.579
n = 4,889
RMSE 29.8
R2 0.770
n = 4,889
Random split
Surrogate [kgCO2e/m2]
Unseen storey counts
Surrogate [kgCO2e/m2]
Full analysis [kgCO2e/m2]
100
200
300
400
500